

A tour of the Ancien Regime: every capability on one synthetic world

The kingdom, and the clerks

The `ancienregime` datasets describe a France that never was: twelve provinces, six occupational classes from `Vagabond` up to `Haute Bourgeoisie`, and a run of head-tax rolls — the capitation of 1750, further rolls in 1770, 1780, and 1790, a dixieme in 1745, and one vingtieme survey in 1780 that asked every household where it had lived ten years before.

None of these records carry a person identifier. They are linked the way historical records are always linked: by clerks matching names, and the clerks make mistakes. When a link fails it does not produce a blank — it produces *somebody else's record*, and every error this package models flows from that fact.

This vignette walks the package's full capability set through that one world, in the order a research project would meet them:

1. **Migration with a sparse outcome** — 240 parishes, too many for the general-purpose machinery, handled by the EM estimators (`misclassifyr_rl_em()`).
2. **A true-transition operator from auxiliary data** — the vingtieme's recalled-residence question plays the role the 1940 census's migration question plays in real work.
3. **The two-step architecture** — a false-link rate estimated where cells are dense, then plugged into the sparse geography.
4. **Local rivals and the stacked model** — clerks searched only their own generalite's registers, so a kingdom-wide error model is wrong in a way that matters (`misclassifyr_rl_em_stacked()`).
5. **A single-link audit** — when the outcome is measured once, a known rival law and an external operator still identify the error rate (`misclassifyr_known_slab()`).
6. **Occupation error beyond linkage, and the same wrong man** — clerk miscoding and dependent link failures, separated by three linked measures (`misclassifyr_traj_em()`).
7. **From the corrected joint to a mobility slope** — the payoff.

Because the world is synthetic, the truth is available — `ancienregime_truth` ships the generating parameters — and every estimate below is put next to it. On real data you do not get that comparison, which is exactly why it is worth seeing once on data where you do.

```
data(ancienregime_parishes)
data(ancienregime_lineages)
data(ancienregime_truth)

head(ancienregime_truth$parish_register, 4)
#>   parish      parish_name province share_1750
#> 1      1 St-Jean-en-Vallee  Alsace 0.001207729
#> 2      2 St-Martin-le-Comtal Alsace 0.001207729
#> 3      3 Ste-Colombe-le-Vieux Alsace 0.007246377
#> 4      4 St-Aubin-la-Chapelle Alsace 0.001207729
```

1. Migration with a sparse outcome

A man's parish in 1750 is observed directly; his parishes in 1770 and 1780 come through links. The naive migration rate — the share of men whose linked 1770 parish differs from their 1750 parish — mixes true moves with false links:

```
d <- ancienregime_parishes
mean(d$parish_1750 != d$parish_1770_linked)
#> [1] 0.3563333
```

Some of that is real migration and some is clerks. With 240 parishes, the tabulation is sparse — most parish pairs are never observed — so we use the EM estimator built for exactly this shape. It wants a tabulated (X, Y1, Y2, n) data frame:

```
tab <- aggregate(list(n = rep(1L, nrow(d))),
                  by = list(X = d$parish_1750,
                           Y1 = d$parish_1770_linked,
                           Y2 = d$parish_1780_linked), FUN = sum)
nrow(tab) # observed cells, out of 240^3 possible
#> [1] 19747
```

2. The vingtieme supplies the true-transition operator

There is a subtlety before we fit. The 1780 measure is *thirty years* after the base record, and men genuinely move in thirty years. A model that treats any 1750-vs-1780 disagreement as linkage error will call true migration a false link. The fix is a **true-transition operator**: tell the model how much genuine movement to expect, and let the false-link rate absorb only the rest.

Where does an operator come from? From auxiliary data the rolls themselves supply. The vingtieme of 1780 asked each household where it resided in 1770 — so the full roll, no linkage involved, yields the decade-long parish-to-parish flow:

```
ving <- ancienregime_truth$vingtieme_flows
head(ving, 3)
#>   j l      t    n
#> 1 1 1 0.834426230 509
#> 2 1 2 0.003278689   2
#> 3 1 3 0.013114754   8
```

This is the toy counterpart of using the 1940 census’s five-year migration question to discipline the operator in real work. We pass it as T2, so the second measure’s model becomes “a correct link observes a draw from true migration; a failed link draws from the rival law.”

```
fit_u <- misclassify_r1_em(tab, J = 240, K = 240,
                          T2 = ving[, c("j", "l", "t")],
                          maxit = 400, tol = 1e-9)
round(fit_u$alpha, 4)
#> alpha1 alpha2
#> 0.0132 0.0236
ancienregime_truth$parishes$alpha
#> [1] 0.07 0.12
```

The fit converges — and it is **badly wrong**, understating both false-link rates several times over. That failure is deliberate, and it is the next lesson.

3. Local rivals: why the kingdom-wide model fails

The model just fitted assumes a failed link draws from one kingdom-wide rival law. But the intendants’ clerks searched only their own generalite’s registers, so a failed link draws from the man’s **birth province’s** roll. A rival from your own province often lands near home — sometimes in your very parish — and the kingdom-wide model reads those as correct links. The estimated rate collapses.

The remedy is to condition the error model the way the linking process was actually run: split by birth

province, let each cell carry its own rival law, and share the false-link rates across cells — which is what `misclassifyr_rl_em_stacked()` does.

```
tabs <- lapply(split(d, d$province_birth), function(z)
  aggregate(list(n = rep(1L, nrow(z))),
    by = list(X = z$parish_1750,
      Y1 = z$parish_1770_linked,
      Y2 = z$parish_1780_linked), FUN = sum))

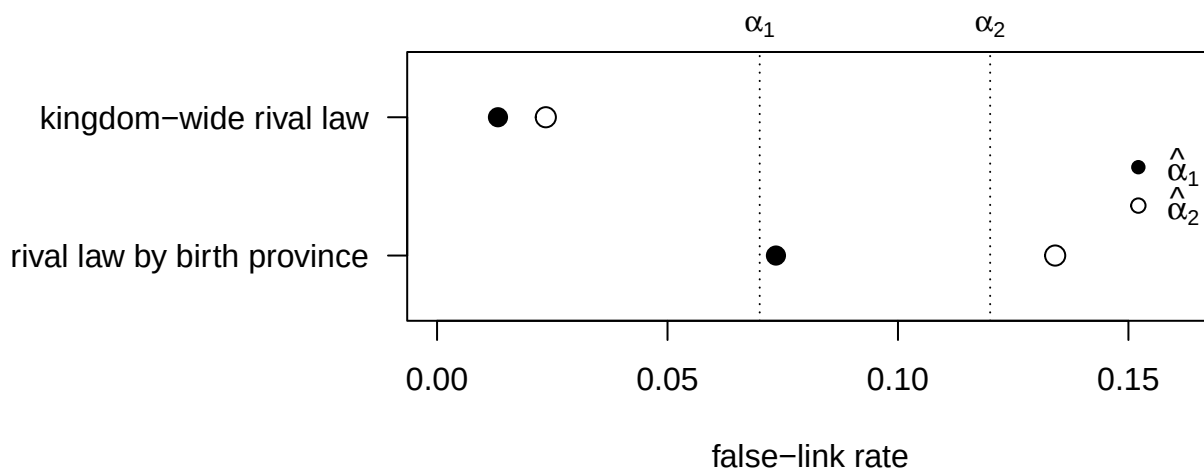
fit_s <- misclassifyr_rl_em_stacked(tabs, J = 240, K = 240,
  T2 = ving[, c("j", "l", "t")],
  maxit = 400, tol = 1e-9)

round(fit_s$alpha, 4)
#> alpha1 alpha2
#> 0.0735 0.1341
ancienregime_truth$parishes$alpha
#> [1] 0.07 0.12
```

Same data, same estimator family, right conditioning — and the truth comes back. The general rule: **condition the rival law on whatever the linking keys condition on**. In real work the keys are name, birthplace, and age; a rival pool that ignores them is too diffuse, spurious draws look like correct links, and the error rate is understated. On this synthetic world the understatement is a factor of five; in real linked census data the same mechanism is why unconditional estimates should be treated as lower bounds.

The corrected migration matrix itself is `fit_s$cells`, one Π_i per province — sparse, support-restricted, and free of the false-link smear.

The section in one picture — same data, same estimator family, only the conditioning of the rival law differs:



The dotted verticals are the true rates; the diffuse rival law puts both estimates near zero, and the conditioned one puts both on the lines.

4. The two-step architecture for very sparse outcomes

With hundreds of categories, even the stacked model can leave Π_i data-hungry. The two-step architecture estimates the false-link rates where cells are dense — at a coarser aggregation — and then holds them fixed while estimating the fine-geography joint. Provinces are our coarse level:

```
reg <- ancienregime_truth$parish_register
to_prov <- match(reg$province, ancienregime_truth$provinces) # parish -> province code
tab_prov <- aggregate(list(n = tab$n),
```

```

      by = list(X = to_prov[tab$X],
                Y1 = to_prov[tab$Y1],
                Y2 = to_prov[tab$Y2]), FUN = sum)

# the operator aggregates the same way
Tm <- matrix(0, 240, 240)
Tm[cbind(ving$j, ving$l)] <- ving$t
Tp <- rowsum(t(rowsum(Tm, to_prov)), to_prov)      # flows province -> province
Tp <- Tp / rowSums(Tp)
T2_prov <- data.frame(j = rep(seq_len(12), each = 12),
                      l = rep(seq_len(12), times = 12),
                      t = as.vector(t(Tp)))

fit_coarse <- misclassif_r_lem(tab_prov, J = 12, K = 12, T2 = T2_prov,
                              maxit = 400, tol = 1e-9)
round(fit_coarse$alpha, 4)
#> alpha1 alpha2
#> 0.0075 0.0017

```

The coarse fit collapses almost to zero — *worse* than the parish-level fit, not better. The reason is worth a moment: a failed link draws from the man's birth province, and most men still live in their birth province, so at province resolution the rival lands on the **same aggregated value as the truth** and the false link becomes literally invisible. Coarsening below the resolution of the rival's locality does not dilute the problem; it erases the evidence. **Coarsening is not conditioning.** The honest coarse step in this world is the stacked fit of section 3 — dense enough cells *and* the right conditioning — and in real work it is whatever level achieves both. With the rates in hand, the fine-geography second step is one argument:

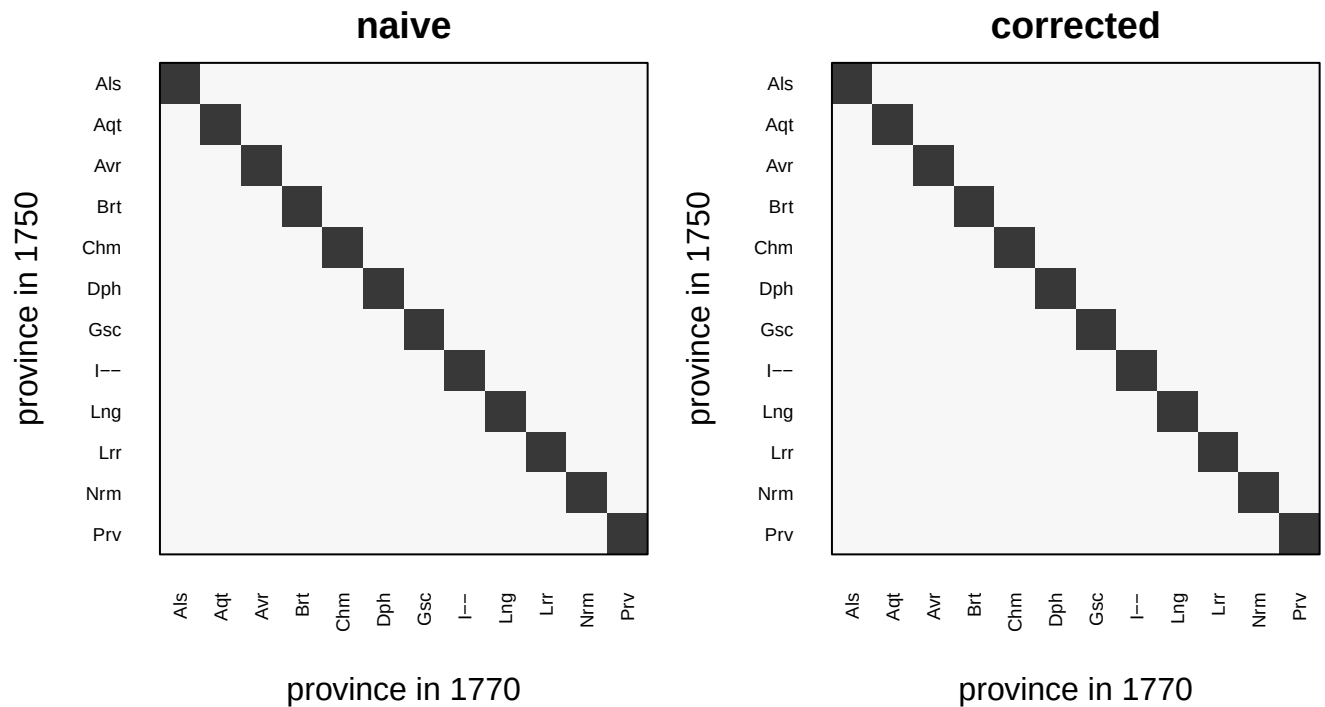
```

fit_fine <- misclassif_r_lem(tab, J = 240, K = 240,
                            T2 = ving[, c("j", "l", "t")],
                            alpha_fixed = fit_s$alpha)

```

(Not run here only to keep the vignette quick; it converges in a few seconds and its Π matches section 3's.)

Since the province objects are now on hand, this is the place to draw the deliverable. The corrected per-province joints of section 3 aggregate to a province-to-province flow matrix; beside the naive tabulation, the correction visibly strips off-diagonal mass — the spurious moves — while leaving the flows that are real:



Rows are 1750 provinces, columns 1770, both row-normalized; the population-weighted share off the diagonal falls from 0.054 in the naive tabulation to 0.057 after correction.

5. The single-link audit: a known rival law

Everything so far used two linked measures. Sometimes there is only one — but if the rival law is *known* rather than estimated, one measure still identifies the false-link rate. In this world the rolls are public: the intendant of Bretagne can put his linked 1750-to-1780 sample next to the full 1780 roll of his own generalite, whose margin is exactly the law his clerks' failed links drew from.

```
own <- reg$parish[reg$province == "Bretagne"]
db <- d[d$province_birth == "Bretagne", ]
tab1 <- aggregate(list(n = rep(1L, nrow(db))),
                  by = list(x = db$parish_1750,
                           y = db$parish_1780_linked), FUN = sum)

# three decades of true migration, from the vingtieme operator composed
T3 <- Tm %*% Tm %*% Tm
Tc <- data.frame(x = rep(seq_len(240), each = 240),
                 y = rep(seq_len(240), times = 240),
                 t = as.vector(t(T3)))
Tc <- Tc[Tc$t > 1e-9, ]

rho_bretagne <- ancienregime_truth$parishes$margin_1780
rho_bretagne[-own] <- 0
rho_bretagne <- rho_bretagne / sum(rho_bretagne)

fit_1 <- misclassifyr_known_slab(tab1, T_cond = Tc, rho = rho_bretagne)
round(fit_1$alpha, 3)
#> [1] 0.141
ancienregime_truth$parishes$alpha[2]
#> [1] 0.12
```

Close, and the gap is instructive: the operator here is the *vingtieme*'s flow table composed three times, sampling noise and all, and the single-link design leans on the operator harder than the two-link design does — there is no second measure to share the burden. Report single-link estimates with an operator-sensitivity band (refit under a perturbed operator) rather than as a point.

This is the design to reach for when a census asks its own migration question: the operator comes from the question, the rival law from the full count, and the single link's error rate falls out. It is also a useful *audit* of the two-link estimates, since it rests on different data.

6. Occupation error beyond linkage, and the same wrong man

Now the harder problem, and the reason the package exists. Turn to the lineages: fathers observed in 1750 (and, through a link, in the 1745 *dixieme*), sons linked into 1770, 1780, and 1790. The estimand is the transmission matrix — father's class to son's class — and *two* error processes stand between the rolls and it:

- **The clerk's pen.** Occupations are miscoded to an adjacent rung with probability μ , on every record, linked or not. A *Journalier* written down as a *Metayer* is not a linkage error, and no linkage model can absorb it correctly.
- **The same wrong man.** When a son's links fail, they do not fail independently: with probability π_{shared} every failed link lands on the *same* rival — a real person, whose own career follows the same decade-to-decade dynamics. Two men confusable in 1770 are confusable in 1780.

Start with what goes wrong if you ignore both. Fit the plain record-linkage model on the first two measures of the son:

```
occ <- ancienregime_truth$occupations
lin <- ancienregime_lineages
code <- function(v) match(v, occ)

tab2 <- aggregate(list(n = rep(1L, nrow(lin))),
                    by = list(X = code(lin$father_occupation_1750),
                              Y1 = code(lin$son_occupation_1770_linked),
                              Y2 = code(lin$son_occupation_1780_linked)),
                    FUN = sum)
fit_2 <- misclassify_r1_em(tab2, J = 6, K = 6, maxit = 500)
round(fit_2$alpha, 3)
#> alpha1 alpha2
#> 0.149 0.287
ancienregime_truth$lineages$alpha[1:2]
#> [1] 0.1023 0.1305
```

Both rates are wrong, and the second is wrong by more than a factor of two. The diagnosis is worth internalizing, because it is the deepest lesson this package has to teach: **the false-link rate is the residual of your model of true change.** The two-measure model here has no occupational dynamics — it expects a correctly-linked son to report the *same* class in 1780 as in 1770 — so every genuine career move between the two rolls, and every stroke of the clerk's pen, has exactly one parameter left to land in. The rate is not measuring linkage any more; it is absorbing everything the model cannot represent. (And the dependence is invisible on top of that: with two measures, no feature of the data distinguishes independent failures from same-rival failures.)

The trajectory estimator fixes all three deficiencies at once: a third linked measure (which identifies the dependence), a transition operator T estimated inside the model (which carries true career moves), and a measurement kernel (which carries the clerk).

```
tab3 <- aggregate(list(n = rep(1L, nrow(lin))),
                    by = list(Y1 = code(lin$son_occupation_1770_linked),
                              Y2 = code(lin$son_occupation_1780_linked),
```

```

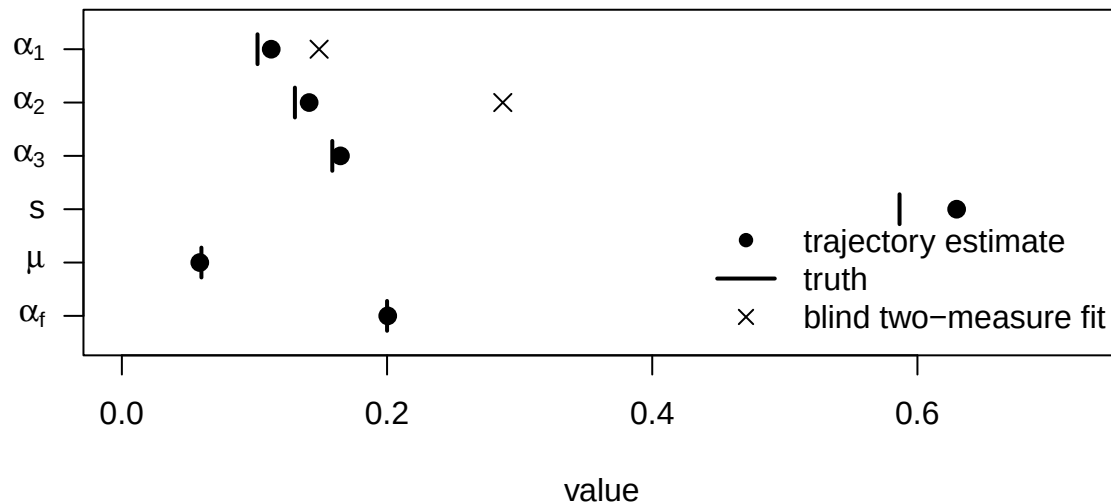
Y3 = code(lin$son_occupation_1790_linked),
X  = code(lin$father_occupation_1750),
Xf = code(lin$father_occupation_1745_linked)),
FUN = sum)

tr <- ancienregime_truth$lineages
fit_3 <- misclassify_traj_em(
  tab3, J = 6,
  rho   = tr$occupation_slab, # the latent slab failed links draw from
  rho_f = tr$father_margin,  # the father's link's rival law
  kernel = tr$kernel,        # the clerk's adjacent-rung confusion
  dstep = c(1, 1),           # one decade between consecutive measures
  dstep_f = 1,               # the dixieme sits one step before the base
  maxit = 800)

rbind(estimate = round(c(fit_3$alpha, s = fit_3$s, mu = fit_3$mu[1],
                        alpha_f = fit_3$alpha_f), 3),
      truth     = round(c(tr$alpha, s = tr$s, mu = tr$mu,
                        alpha_f = tr$alpha_f), 3))
#>      alpha1 alpha2 alpha3      s      mu alpha_f
#> estimate 0.113 0.141 0.165 0.630 0.059      0.2
#> truth     0.102 0.130 0.159 0.587 0.060      0.2

```

Every parameter comes back: the three marginal false-link rates, the share s of failures that hit the same rival, the clerk's μ , and the father's own link rate α_f — the last identified because the father, too, is measured twice. Note what the estimator was *given*: the rival laws (ρ , ρ_f) as data, derived here from the world's registers, exactly as real work derives them from full-count margins — and note the docstring's warning that with a `kernel` these are laws over the *latent* class.



The vertical bars are the truth, the dots the trajectory estimates, and the crosses the two-measure fit of the previous chunk — sitting far off the bars on the two rates it can see, having absorbed the clerk and the career dynamics it cannot.

The two practical warnings from the estimator's documentation bear repeating. The (α, s) block sits on a likelihood plateau, so use the tight default tolerance and, on real data, multiple starts. And a fitted s at meso resolution is a *lower bound* on the dependence: rivals are selected for confusability, and confusable men do similar work, so a same-rival failure is often invisible in the occupation alphabet.

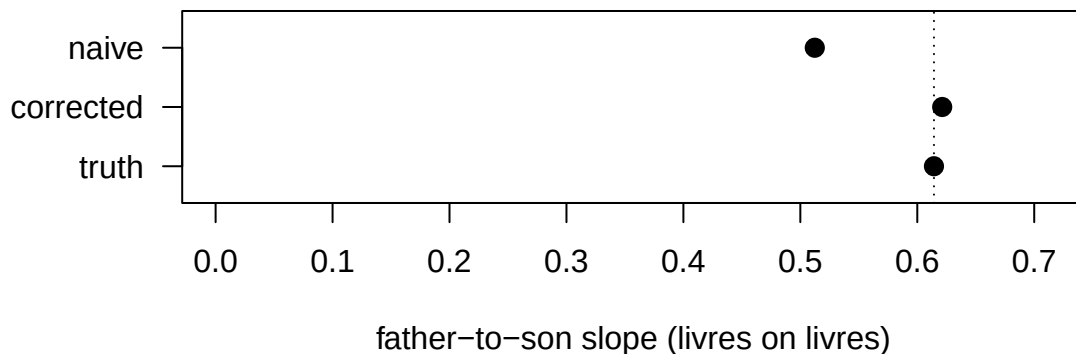
7. The payoff: a corrected mobility slope

The object of interest was never `alpha` — it was the transmission of standing from father to son. Attach a livre-denominated income score to each class and compare three slopes: the naive one from the linked records, the corrected one from `fit_3$Pi`, and the truth.

```
scores <- c(10, 25, 30, 45, 90, 250) # livres, one per class
slope_from_joint <- function(P) {
  P <- P / sum(P)
  fx <- rowSums(P); fy <- colSums(P)
  mx <- sum(scores * fx); my <- sum(scores * fy)
  cxy <- sum(outer(scores - mx, scores - my) * P)
  vx <- sum((scores - mx)^2 * fx)
  cxy / vx
}

Pi_true <- tr$Pi * tr$father_margin # joint from the truth
Pi_naive <- with(lin, table(factor(code(father_occupation_1750), 1:6),
                              factor(code(son_occupation_1770_linked), 1:6))) |>
  prop.table() |> unclass()

c(naive = slope_from_joint(Pi_naive),
  corrected = slope_from_joint(fit_3$Pi),
  truth = slope_from_joint(Pi_true)) |> round(3)
#>      naive corrected      truth
#>      0.512      0.621      0.614
```



The naive slope is attenuated — false links put strangers' sons under fathers they do not belong to, and the clerk's noise attenuates further — and the corrected slope sits on the truth. For delta-method standard errors and Monte Carlo confidence sets on slopes like this one, see `se_beta_deltamethod()`, `Pi_to_beta()`, and `vignette("inference")`.

8. One error process, two outcomes

A link fails or holds once, whatever is read off the linked record. In real work the same crosswalk carries a man's county *and* his occupation, and fitting the two outcomes separately invites each fit to park its own outcome's problems in the shared quantity — the false-link rate. `misclassify_traj_em_shared()` estimates outcome *views* jointly with the linkage component forced equal: one (`alpha`, `s`) process, one `Pi`, `T`, and measurement layer per view.

Here the second view is the intendant's digest: the six classes rolled up into three orders, with the kernel and the rival law aggregated to match (weighted by the sons' latent class margin, as any honest aggregation of a conditional law must be):


```

grp <- c(1, 1, 2, 2, 3, 3) # six classes into three orders
m_son <- as.numeric(tr$father_margin %*% tr$Pi) # sons' latent margin

aggC <- function(M, w) { # conditional-law aggregation
  A <- rowsum(M * w, grp); A <- t(rowsum(t(A), grp)); A / rowSums(A)
}

kern3 <- aggC(tr$kernel, m_son) # the kernel is a conditional law
rho3 <- as.numeric(rowsum(tr$occupation_slab, grp)) # marginal laws just sum
rho_f3 <- as.numeric(rowsum(tr$father_margin, grp))

g <- function(v) grp[v]
tab3c <- aggregate(list(n = tab3$n),
  by = list(Y1 = g(tab3$Y1), Y2 = g(tab3$Y2),
            Y3 = g(tab3$Y3), X = g(tab3$X),
            Xf = g(tab3$Xf)), FUN = sum)

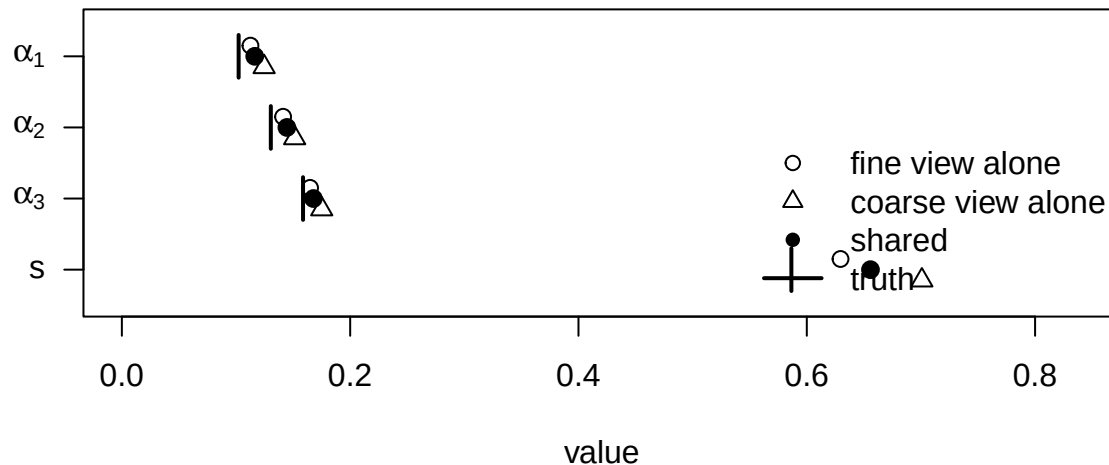
fit_c <- misclassify_traj_em(tab3c, J = 3, rho = rho3, rho_f = rho_f3,
  kernel = kern3, dstep = c(1, 1),
  dstep_f = 1, maxit = 800)

fit_sh <- misclassify_traj_em_shared(
  views = list(
    classes = list(tab = tab3, J = 6, rho = tr$occupation_slab,
      rho_f = tr$father_margin, kernel = tr$kernel,
      dstep = c(1, 1), dstep_f = 1),
    orders = list(tab = tab3c, J = 3, rho = rho3, rho_f = rho_f3,
      kernel = kern3, dstep = c(1, 1), dstep_f = 1)),
  maxit = 400)

rbind(`fine alone` = round(c(fit_3$alpha, s = fit_3$s), 3),
  `coarse alone` = round(c(fit_c$alpha, s = fit_c$s), 3),
  shared = round(c(fit_sh$alpha, s = fit_sh$s), 3),
  truth = round(c(tr$alpha, s = tr$s), 3))

#>           alpha1 alpha2 alpha3 s
#> fine alone  0.113  0.141  0.165 0.630
#> coarse alone 0.125  0.151  0.175 0.701
#> shared      0.116  0.145  0.168 0.656
#> truth       0.102  0.130  0.159 0.587

```



Both views alone are serviceable here because both were handed their kernels; the shared fit agrees with them and with the truth, on one error process instead of two. The force of the constraint shows when a view's measurement layer is *incomplete*: whatever that view can no longer park in **alpha** is pushed back into its own emission structure, where a misspecification is visible, rather than into the linkage rate, where it is not. In real work the two views are county and occupation read through the same crosswalk, and the shared fit is the answer to “the two designs disagree about the same edge.”

What to take back to real data

Each section above is one lesson, and they compound:

1. **Sparse outcomes need the EM estimators**, and a tabulation is all they ask for.
2. **Bring an operator for true change** — from an auxiliary question, an external survey, anything not built on the links themselves — or the false-link rate will eat true change.
3. **Estimate the error rate where cells are dense**, then fix it where they are sparse.
4. **Condition the rival law on the linking keys**. An unconditional rival pool is too diffuse, and the error rate collapses toward zero. Treat unconditional estimates as lower bounds.
5. **A known rival law rescues the single-link design**, and audits the others.
6. **Two measures cannot see dependent failures, and a linkage-only model turns true dynamics and miscoding into “false links.”** The false-link rate is the residual of your model of true change; before trusting a correction, ask whether that model is rich enough that the rate is not quietly standing in for it.
7. **The corrected joint, not the error rate, is the deliverable** — and it feeds straight into downstream slopes with inference.
8. **One error process per crosswalk, however many outcomes you read through it**. Fit the views jointly with the linkage component shared, and let each outcome's own measurement layer carry its own noise.

All of it, on a kingdom that never existed, from rolls no clerk ever botched.